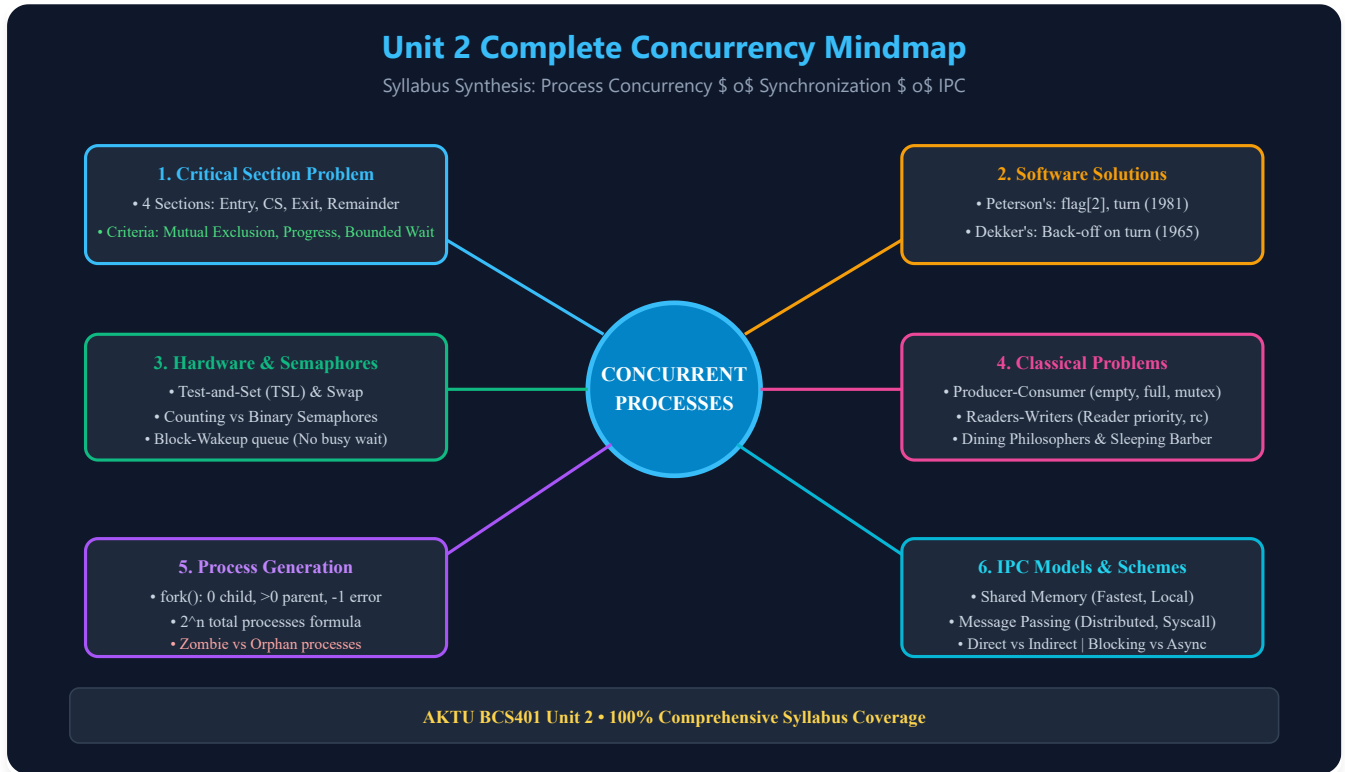


12. Unit 2: AKTU PYQs & Interview Cheatsheet

Syllabus Coverage: Gateway Classes Slides 40, 60, 92, 106, 120, 136 | Solved AKTU Question Bank (2014–2023), Top Interview Traps, Concurrency Mindmap.



12.1 High-Yield Solved Exam Questions

- **Critical Section Criteria:** Mutual Exclusion (Primary), Progress (Primary), Bounded Waiting (Secondary), Speed Neutrality.
- **Peterson's Algorithm:** Proves mutual exclusion via single integer `turn`. Bounded wait $k=1$.
- **Overcoming Busy Waiting:** Use `block()` and `wakeup()` queue-based semaphore implementation.
- **Producer-Consumer Trap:** Always place counting semaphore (`empty/full`) before mutex to prevent deadlock.
- **Process Tree Formula:** n consecutive forks yield 2^n total processes and $2^n - 1$ child processes.
- **Zombie vs Orphan:** Zombie = Dead child waiting for parent's `wait()`; Orphan = Running child adopted by PID 1.